

**Entrepreneurship Development and Innovation Institute
(EDII-TN)**

Innovation Voucher Programme

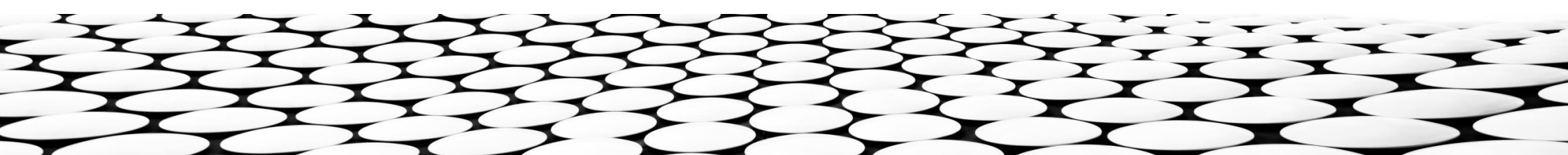
**SHA SMART BRIDGE-X: A UNIFIED MIDDLEWARE
HARDWARE FRAMEWORK FOR SECURE ORCHESTRATION
OF UAV FUNCTIONAL INTERFACES**

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TTS No.3981

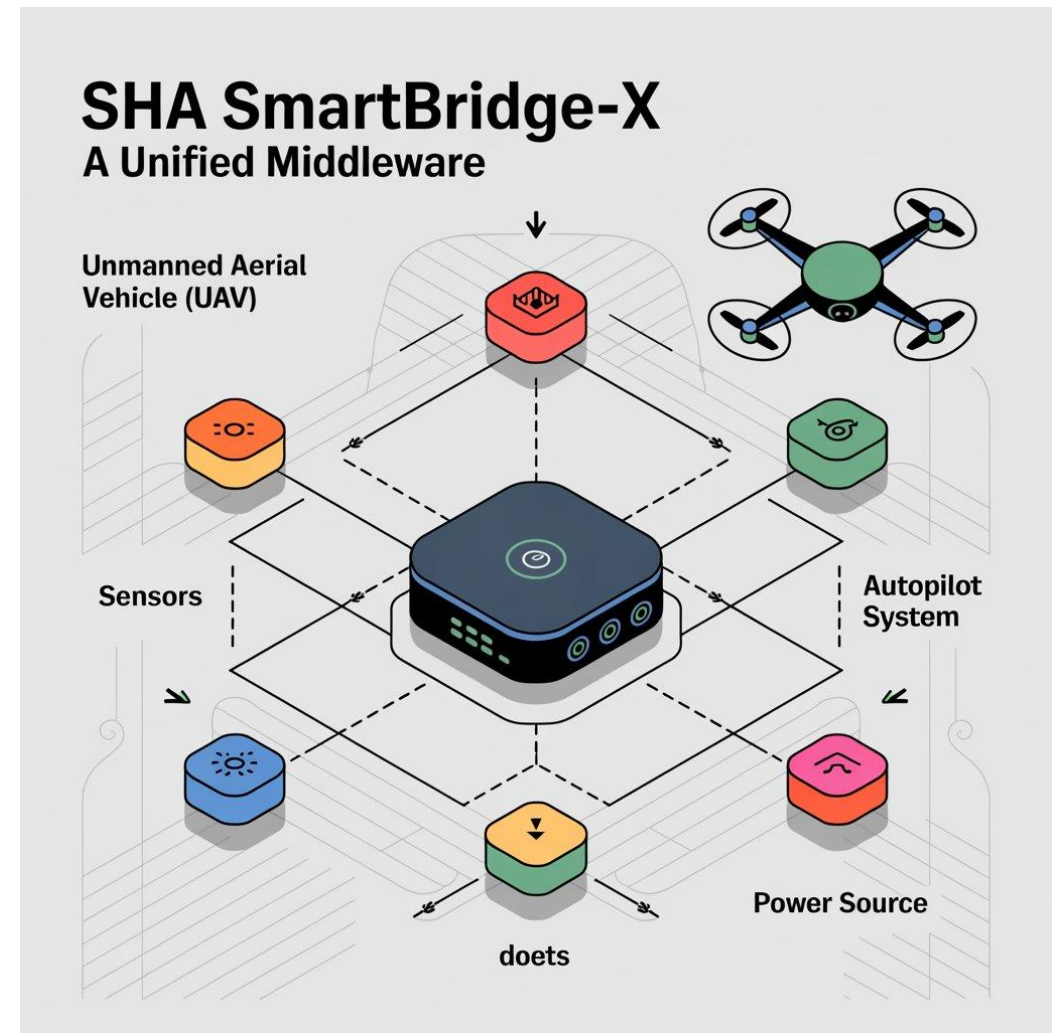
Assistant Professor, CSE(CS)

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PRODUCT OVERVIEW

- A unified, secure hardware middleware framework designed to simplify and safeguard UAV system integrations by enabling streamlined communication across sensors, telemetry, autopilot, and power systems.



PROBLEM STATEMENT

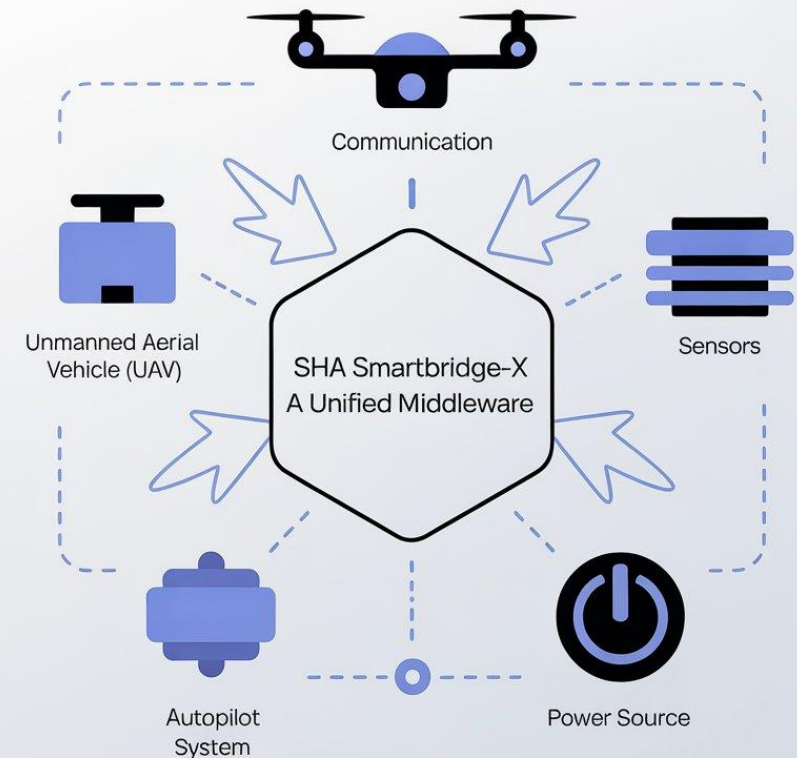
PROBLEM (P)

- Modern UAV systems face integration challenges due to multiple independent modules causing inefficiencies, unreliability, and increased complexity.

PROPOSED SOLUTION

INTERVENTION (I)

- SHA Smart Bridge-X is an indigenous middleware framework with unified IO management, tamper detection, and intelligent power control improving reliability and operational security.



PRODUCT & SERVICES

DESCRIPTION

- Hardware: Raspberry Pi CM4-based core with unified IO ports (USB, UART, SPI, I2C)
- Software: Secure orchestration logic embedded in middleware
- Vision: CSI/USB camera support
- Communication: Mesh Radio / 5G / Wi-Fi module integration
- Power Management: BEC-based regulated output

INNOVATION TYPE

NEW INVENTION – WHY?

- Provides a secure, consolidated control platform replacing fragmented breakout board-based integration methods.

PROJECT SCOPE

SCOPE

- AV-level orchestration (control, data, comms, sensors)
- Applicable to academic research, defence, agritech, and surveillance
- Scalable for both single and multi-drone systems

PROJECT NEED

NEED

- High demand for reliable UAV integration systems
- Rising UAV applications in defense, agriculture, and delivery systems
- Compliance with national drone security and scalability standards

COMPETITIVE & TECHNICAL ADVANTAGE

COMPARISON (C)

- Existing: Loosely connected modules with higher fault probability
- SHA Smart Bridge-X: Single orchestration unit with tamper detection and unified access

ADVANTAGES

- Faster integration
- Fewer points of failure
- Built-in security and compatibility features

SOCIAL & ENVIRONMENTAL IMPACT

IMPACT

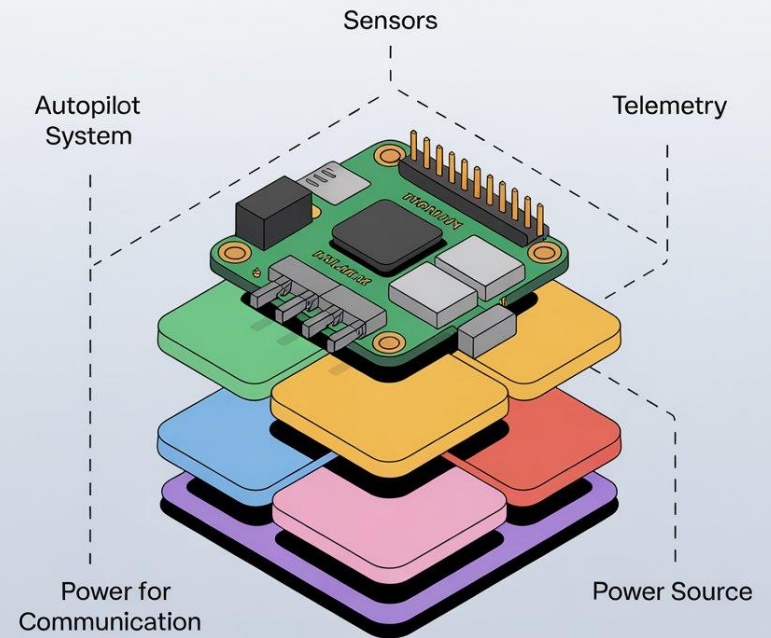
- Reduces e-waste via integrated design
- Enables drone based Agri/Defence related surveillance, reducing chemical overuse
- Supports disaster response and monitoring

COLLABORATIONS

POTENTIAL OR ONGOING COLLABORATIONS

- Academic UAV labs
- Defense research startups
- Agri-tech UAV manufacturers

SHA Smartbridge-X: A Unified Middleware



TANGIBLE BENEFITS

OUTCOMES

- 50% reduction in integration time
- 30% decrease in power failure issues
- Enhanced UAV reliability in mission-critical environments

CUSTOMER PRICE ADVANTAGE

COST SAVINGS

- Eliminates need for multiple expansion boards
- Reduced technician hours for wiring/debugging
- Lower long-term maintenance costs

PROJECT COST

ESTIMATED BUDGET

Budget Split-up – Initial Voucher A (₹2,00,000):

- **Essential Hardware – ₹80,000**

Includes Raspberry Pi CM4, sensors (GPS, IMU, barometer), and a basic telemetry unit.

- **Prototyping Materials – ₹30,000**

Covers PCB printing, connectors, power regulation components, and basic casing materials.

- **Embedded Development Setup – ₹30,000**

For basic firmware development and middleware interface coding (UART/SPI/I2C integration).

- **Testing & Calibration – ₹20,000**

Used for internal lab bench testing setup, basic tools, and calibration.

- **Documentation – ₹10,000**

Supports initial design documentation, architectural diagrams, and circuit drawings.

- **Contingency – ₹30,000**

Reserved for part availability issues, small upgrades, or unexpected rework.

Total Estimated Budget (Voucher A): ₹2,00,000

TIMELINE - MILESTONES



SUMMARY

- *“SHA Smart Bridge-X delivers a secure, unified control layer for next-gen UAVs—boosting reliability and simplifying integration”.*

Call to Action:

- *Support the development of indigenous drone technologies.*

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THANK YOU